1. Determine the following limits.

   (a) $\lim_{x\to-\infty} -2x^4 =$

   (b) $\lim_{x\to\infty} 2x^{-4} =$

   (c) $\lim_{x\to-\infty} 5\cdot\dfrac{3}{x} =$

   (d) $\lim_{x\to-\infty} -2x^4 =$

2. Write the function $\dfrac{-3(x^2)^4}{5}$ as a power function of the form $kx^p$. Enter the values of $k$ and $p$ in the spaces provided.

   $k =$ _____ $\qquad\qquad p =$ _____

3. Circle all of the polynomials.

   (a) $f(x) = 6^x + 4$

   (b) $f(x) = x^6 + 4$

   (c) $f(x) = 6x^{3.7} + 2$

   (d) $f(x) = 6x^2 + 3.7$

   (e) $f(x) = 2x^{-6} + 3.7$

4. Determine the degree and the long-run behavior of each polynomial.

   (a) $f(x) = 9x^6 - 3961x^2 + 6$

   degree of $f$ : _____ $\qquad \lim_{x\to\infty} f(x) =$ _____ $\qquad \lim_{x\to-\infty} f(x) =$ _____

   (b) $f(x) = 8x^3 + \frac{6x^4}{x^{-6}} - 7x^5 + 1$

   degree of $f$ : _____ $\qquad \lim_{x\to\infty} f(x) =$ _____ $\qquad \lim_{x\to-\infty} f(x) =$ _____

5. Determine the zeros of the following polynomials. Do not approximate.

   (a) $f(x) = x^3(7 - 2x^9)$

   (b) $f(x) = (5 - 2x)(3x - 2)^2$

6. An open-top box is to be constructed from a 6 inch by 14 inch rectangular sheet of tin by cutting out squares of equal size at each corner, then folding up the resulting flaps. Let $x$ denote the length of the side of each cut-out square. Assume negligible thickness.

   (a) Find a formula for the volume, $V$, of the box as a function of $x$.

   (b) For what values of $x$ does your formula make sense in the context of the problem?

   (c) Create a sign chart and sketch the graph of $V(x)$.